



Institute of Information Technology,

University of Dhaka

Software Requirements Specification and Analysis

Software Project Lab - III [SE-801]

Project Name: AcoustiCare

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Supervisor's Signature

Letter of Transmittal

30th November, 2024

BSSE 4th Year Exam Committee

Institute of Information Technology

University of Dhaka

Subject: Technical Report Submission - AcoustiCare: Acoustic Analysis and Prediction of Type 2 Diabetes Using Smartphone-Recorded Voice Data

Sir,

I am submitting the technical report for our Software Project Lab 3, "**AcoustiCare**: Acoustic Analysis and Prediction of Type 2 Diabetes Using Smartphone-Recorded Voice Data". This report provides a detailed overview of the project's technical aspects, including its design, implementation, and testing phases.

This report comprehensively covers the project's technical details, including design, implementation, and testing phases. While I've tried to do my best, I welcome your feedback for improvement.

Thank you for your consideration.

Sincerely,

Rifah Tashfiha Faria

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_____

Supervisor's Signature

Acknowledgment

I am grateful to the Institute of Information Technology for giving me such a tremendous opportunity to work on “”. I would like to convey my tremendous gratitude to my supervisor, Dr. Ahmedul Kabir, Associate Professor, Institute of Information Technology, University of Dhaka, for providing me guidelines about how I can prepare this analysis document. Lastly, I would like to thank my classmates. They have always been helpful and provided valuable insights from time to time.

Abstract

This document outlines the software requirements, specifications, architectural design, and testing of "**AcoustiCare**: Acoustic Analysis and Prediction of Type 2 Diabetes Using Smartphone-Recorded Voice Data". The project collects and studies research articles on Type 2 Diabetes Mellitus (T2DM) and its correlation with acoustic features of voice. It leverages internet resources to gather relevant articles, extract key insights, and analyze specific parameters related to the impact of T2DM on vocal characteristics.

The application also highlights essential keywords and provides various visualizations to enhance user understanding and interaction. These insights can assist individuals, healthcare professionals, diagnostic centers, and medical organizations in understanding and utilizing the application for further research or diagnostic purposes.

Motivation

The prevalence of Type 2 Diabetes Mellitus (T2DM) has been rapidly increasing worldwide, necessitating innovative and accessible methods for early detection and monitoring. Traditional diagnostic procedures for T2DM typically require invasive tests, specialized equipment, and frequent clinical visits, which may be inaccessible or inconvenient for many individuals. Addressing this gap, AcoustiCare carries the potential of non-invasive, smartphone-based acoustic analysis to offer a novel solution for T2DM prediction.

This application not only empowers users with a convenient and cost-effective way to monitor their health but also democratizes access to early risk assessment for T2DM. It supports individuals in gaining greater awareness of their health status and enables them to make informed decisions without requiring immediate clinical intervention. Ultimately, AcoustiCare represents a significant advancement in mobile health technologies, integrating sophisticated machine learning and acoustic analysis to contribute meaningfully to preventive healthcare and the fight against T2DM.

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1. Introduction

This chapter provides an overview of the software requirements and specifications for AcoustiCare. It outlines the purpose, scope, and target audience of the project, offering a comprehensive foundation for development.

1.1 Purpose

The objectives of this document are as follows:

- To provide a detailed outline of the software requirements for AcoustiCare, a mobile application designed for acoustic analysis and Type 2 Diabetes Mellitus (T2DM) prediction using voice data.
- To clearly define functional and non-functional requirements, ensuring a shared understanding among all stakeholders.
- To establish a solid baseline for the development process, aligning expectations and facilitating effective project execution.

1.2 Scope

The scope of AcoustiCare is as follows:

- The application will be a mobile platform.
- Users will be able to create profiles and record their voices for acoustic analysis.
- Real-time analysis and feedback will be provided based on the acoustic features of the recorded voice.
- The app will offer data visualization for tracking T2DM trends over time.
- Features like profile switching and data persistence will ensure a seamless and personalized user experience.

1.3 Project Timeline

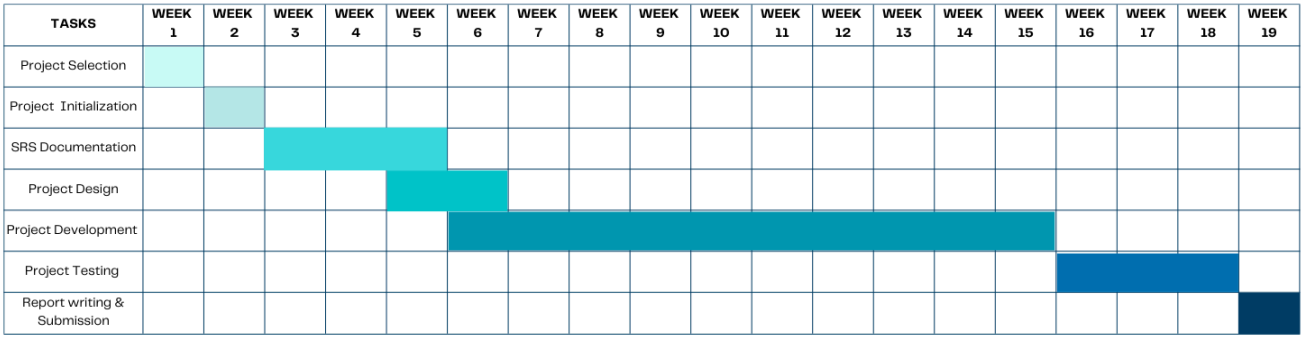


Figure-1: Timeline for AcoustiCare

1.4 Purpose of Document

The purposes of this document are as follows:

- Identify and analyze the requirements for AcoustiCare.
- Design the architectural framework for the application.
- Develop a robust test plan to ensure application reliability.
- Describe the methodology for feature implementation and testing.
- Minimize development effort by streamlining processes.
- Enhance overall understanding and clarity of project goals and deliverables.

2. Project Description

The following outlines the overview of the core functionalities and objectives of AcoustiCare. The journey to conceptualizing and defining the application began with an extensive review of relevant research literature, particularly the paper by Kaufman, Thommandram, and Fossat, which demonstrated the feasibility of detecting physiological conditions like diabetes through acoustic analysis. This was further enriched through consultations with healthcare professionals, machine learning experts, and potential users, ensuring that the application aligns with practical needs and expectations.

2.1. Core Functionalities

AcoustiCare aims to provide a non-invasive solution for early detection of Type 2 Diabetes Mellitus (T2DM) through acoustic analysis of voice data. The functionalities are grouped into three categories:

2.1.1. Normal Requirements

Voice Recording: AcoustiCare must allow users to record predefined voice phrases multiple times daily using a smartphone microphone.

Feature Extraction: The app must analyze the recorded audio to extract key acoustic features like pitch, jitter, and shimmer using tools such as Praat or similar libraries.

Machine Learning Predictions: Based on the extracted features, the app should predict the risk of T2DM using gender-specific machine learning models.

Real-Time Feedback: Users must receive real-time feedback on their T2DM risk with clear and concise explanations.

Profile Management: Users will be able to use multiple profiles in a single device and they will be able to edit their own profiles.

2.1.2. Expected Requirements

Prediction Accuracy: The machine learning models should exhibit high accuracy in assessing T2DM risk.

User-Friendly Interface: The app must ensure an intuitive and seamless user experience, including easy navigation and voice recording procedures.

2.1.3. Exciting Requirements

Trend Analysis: AcoustiCare could provide users with graphical representations of their T2DM risk trends over time.

Health Recommendations: Personalized health suggestions for lifestyle improvements based on the risk analysis.

Customizable Profiles: Users can maintain individual profiles with health data for better tracking and insights.

2.2. Usage Scenarios

AcoustiCare will serve as a smartphone-based application designed to detect and monitor T2DM risk through acoustic analysis. The usage scenarios are as follows:

2.2.1. Voice Recording

Users will open the app and follow instructions to record predefined phrases using their smartphone's microphone. These recordings will form the basis for acoustic analysis.

2.2.2. Feature Extraction and Analysis

The recorded audio will be processed to extract acoustic features such as pitch, jitter, shimmer, and harmonic noise ratio. These features will then be input into machine learning models trained specifically for T2DM risk prediction, considering gender-specific differences.

2.2.3. Real-Time Results and Insights

Upon completing the analysis, AcoustiCare will provide immediate feedback on the user's risk level. The results will include clear explanations of the prediction and its contributing factors.

2.2.4. Trend Tracking and Reporting

Users can track their risk trends over time via interactive graphs. This feature will help monitor changes and identify patterns, encouraging users to maintain or improve their health.

2.2.5. Personalized Health Suggestions

Based on the analysis and trends, the app will provide actionable recommendations for lifestyle changes, such as dietary adjustments and physical activity plans.

2.2.6. Profile Management

AcoustiCare will allow users to create and manage multiple profiles for individual tracking. The app will automatically load the last active profile, ensuring continuity.

This structured approach ensures that AcoustiCare fulfills its goal of providing an accessible, non-invasive, and user-centric solution for early T2DM detection and health improvement.

3. Scenario-Based Modeling

Scenario-based modeling is a method used in software development to create models that represent system behavior under various real-world conditions. For AcoustiCare, the scenarios focus on user interactions and how the app handles different events, enabling developers to identify potential issues and design solutions effectively.

3.1 Use Case Diagram

A Use Case describes the system behavior under various conditions as the system responds to a request from one of its stakeholders. In fact, a use case diagram is a kind of visualization of the system where an end-user has an idea of a specific feature. It simply describes a story using corresponding actors who perform important roles in the story and makes the story understandable for the users.

Level-0:

Name: AcoustiCare

Primary Actor: User

Goal in Context: To provide users with a basic interface for accessing AcoustiCare's voice-based diabetes risk assessment system.

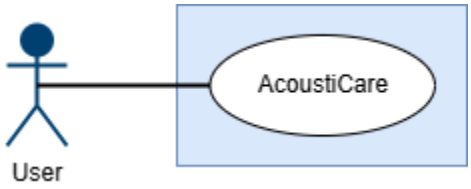


Figure-2: Use Case Diagram L-0

Level-1:

Name: AcoustiCare (Detailed)

Primary Actor: User

Goal in Context: The diagrams show modules of AcoustiCare:

- Managing their profiles and settings
- Recording and analyzing voice samples
- Tracking health trends over time
- Receiving personalized health suggestions
- Managing system notifications

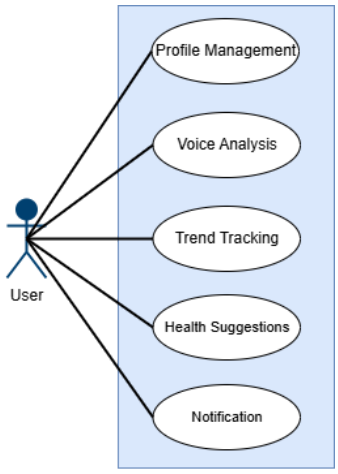


Figure-3: Use Case Diagram L-1

Level-1.1:

Name: Profile Management

Primary Actor: User

Goal in Context: The diagrams shows sub modules of profile management module:

- Creating new accounts through registration
- Accessing existing accounts through login
- Recovering forgotten passwords
- Customizing profile settings
- Switching between multiple user profiles on the same device

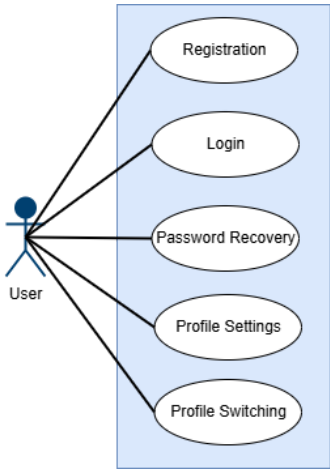


Figure-4: Use Case Diagram L-1.1

Level-1.2:

Name: Voice Analysis

Primary Actor: User

Goal in Context: The diagrams shows sub modules of voice analysis module:

- Recording voice samples
- Extracting relevant acoustic features from recordings
- Analyzing extracted features to assess diabetes risk

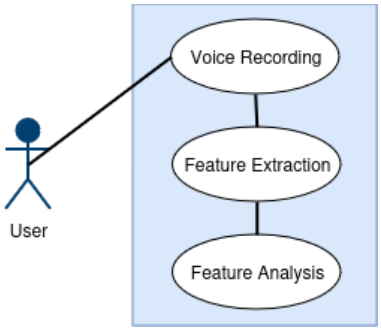


Figure-5: Use Case Diagram L-1.2

4. Data-Based Modeling

Database modeling is a visual representation of a database structure that demonstrates how data is organized within the system. It involves creating entity-relationship diagrams (ERDs) to depict tables (entities), attributes (columns), and their relationships. The process begins by analyzing the system's data requirements, identifying entities and their attributes, and using modeling tools to design ERDs, which serve as the blueprint for database implementation.

After analyzing the data requirements of AcoustiCare, the following entities have been identified:

- VoiceData
- User
- Notifications
- TrendData

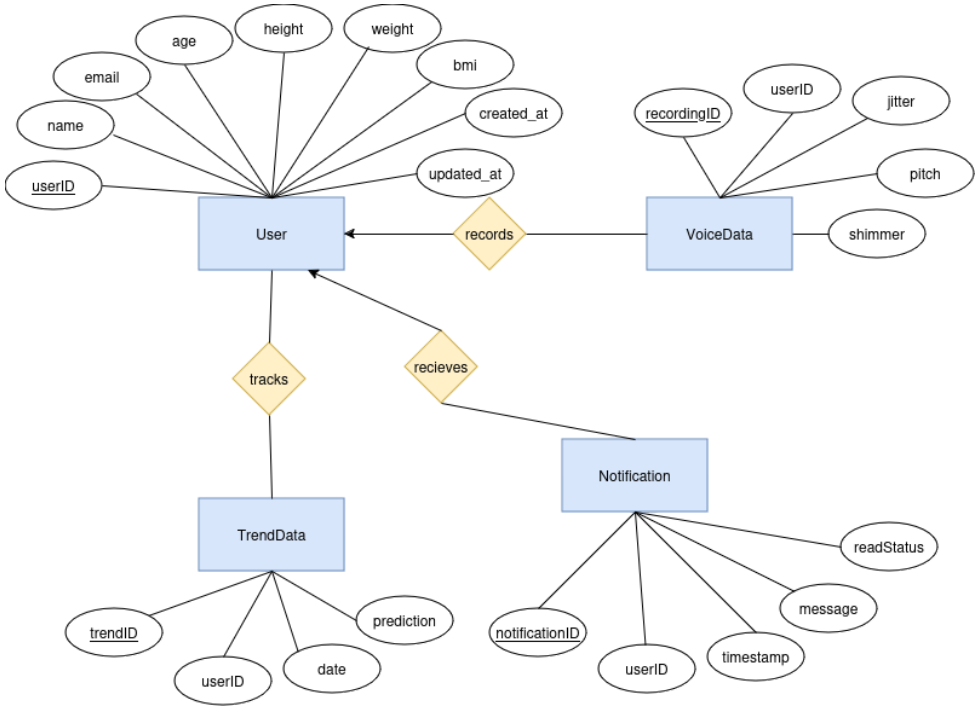


Figure-6: ER-Diagram of AcoustiCare

The schemas derived from the ER-diagram is as follows:

Table-1: Schemas of AcoustiCare

Schema	**Attribute**	**Type**
User	userID	VARCHAR(255) PK
	name	VARCHAR(255)
	email	VARCHAR(255)
	age	INTEGER
	height	FLOAT
	weight	FLOAT
	bmi	FLOAT
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

VoiceData	recordingID	VARCHAR(255) PK
	userID	VARCHAR(255) FK
	recording_timestamp	TIMESTAMP
	file_path	VARCHAR(255)
	pitch	FLOAT
	jitter	FLOAT
	shimmer	FLOAT
TrendData	trendID	VARCHAR(255) PK
	userID	VARCHAR(255) FK
	trend_data	JSON
	timestamp	TIMESTAMP
Notification	notificationIDm	VARCHAR(255) PK
	userID	VARCHAR(255) FK
	notification_type	VARCHAR(255)
	message	TEXT
	timestamp	TIMESTAMP
	read_status	BOOLEAN

5. Class-Based Modeling

This chapter describes the class-based modeling of the AcoustiCare tool. Class-based modeling is a visual representation in software engineering that illustrates the structure of a software system using classes, their attributes, and relationships. It is commonly used in object-oriented design and modeling.

5.1. Analysis Classes

After identifying key entities from the scenario, I filtered them using General Classification (e.g., External entities, Things, Roles) and Selection Criteria (e.g., Retained information, Needed services, Essential requirements). The identified classes for AcoustiCare are:

- User Profile
- Profile Management
- Voice Recording
- Feature Extraction
- Model Training
- Risk Prediction
- Trend Tracking
- Health Suggestions
- Notification

5.2 Class Cards

Table-2: CRC Card of User Profile

User Profile	
Attributes	Methods
• User ID • Name • Email • Age • Gender • Height • Weight • BMI	• +CreateProfile() • +Login() • +UpdateProfile() • +ChangePassword() • +DeleteProfile()
Responsibility	Collaborators
• Manage user profile and account details. • Provide authentication and authorization services. • Allow users to update their information. • Provide a mechanism to delete the user account. • Calculate and store BMI based on user	• Trend Tracking • Health Suggestions • Profile Management

input.	

Table-3: CRC Card of Profile Management

Profile Management	
Attributes	Methods
● Active Profile ID ● Available Profiles (List of Profile IDs)	● +GetActiveProfile() ● +SwitchProfile(profileID) ● +ListAvailableProfiles()
Responsibility	Collaborators
● Manage the active profile in the application. ● Allow users to switch between different profiles. ● Ensure data consistency by loading the correct profile's details, trends, and suggestions.	● User Profile

Table-4: CRC Card of Voice Recording

Voice Recording	
Attributes	Methods
● Recording ID ● User ID ● Recording Timestamp ● Audio File Path	● +StartRecording() ● +SaveRecording() ● +GetRecording() ● +DeleteRecording()
Responsibility	Collaborators
● Enable users to record and store audio data for analysis. ● Retrieve recordings for feature extraction.	● Feature Extraction ● User Profile

Table-5: CRC Card of Feature Extraction

Feature Extraction	
Attributes	Methods
• Recording ID • Pitch • Jitter • Shimmer	• +ExtractFeatures() • +StoreFeatures() • +RetrieveFeatures()
Responsibility	Collaborators
• Extract relevant acoustic features from recorded audio. • Provide data for the Risk Prediction model.	• Voice Recording • Model Training • Risk Prediction

Table-6: CRC Card of Model Training

Model Training	
Attributes	Methods
• Training Data Path • Model Type (e.g., Gender-specific models) • Training Parameters • Model Accuracy	• +TrainModel() • +EvaluateModel() • +SaveModel() • +LoadModel() • -UpdateModel()
Responsibility	Collaborators
• Train machine learning models using acoustic features and labeled data. • Evaluate and improve model accuracy. • Manage the lifecycle of predictive models, including saving and loading trained models.	• Feature Extraction • Risk Prediction

Table-7: CRC Card of Risk Prediction

Risk Prediction	
Attributes	Methods
• Prediction ID • User ID • Risk Level • Timestamp	• +PredictRisk() • +GetPredictionDetails() • +SavePredictionResults()
Responsibility	Collaborators
• Predict T2DM risk using extracted features and trained machine learning models. • Store prediction results for trend tracking.	• Model Train • Trend Tracking

Table-8: CRC Card of Trend Tracking

Trend Tracking	
Attributes	Methods
• User ID • Trend Data • Timestamp	• +AnalyzeTrends() • +GetTrends() • +UpdateTrends()
Responsibility	Collaborators
• Track and visualize T2DM risk trends over time. • Provide data for Health Suggestions.	• Risk Prediction • Health Suggestions

Table-9: CRC Card of Health Suggestion

Health Suggestion	
Attributes	Methods
• User ID • Diet • Exercise • Lifestyle	• +GenerateSuggestions() • +GetSuggestions()
Responsibility	Collaborators
• Provide personalized health recommendations based on user trends. • Help users adopt preventive measures for T2DM.	• Trend Tracking • User Profile

Table-10: CRC Card of Notification

Notification	
Attributes	Methods
• Notification ID • User ID • Notification Type • Notification Message • Timestamp • Read Status	• +CreateNotification() • +SendNotification() • +MarkAsRead(notificationID) • +GetNotifications()
Responsibility	Collaborators
• Generate and manage notifications for users. • Deliver real-time alerts about T2DM risks, trends, or health suggestions. • Allow users to review and manage their notification history.	• Health Suggestions • User Profile

5.3. Class Responsibility and Collaboration Diagram

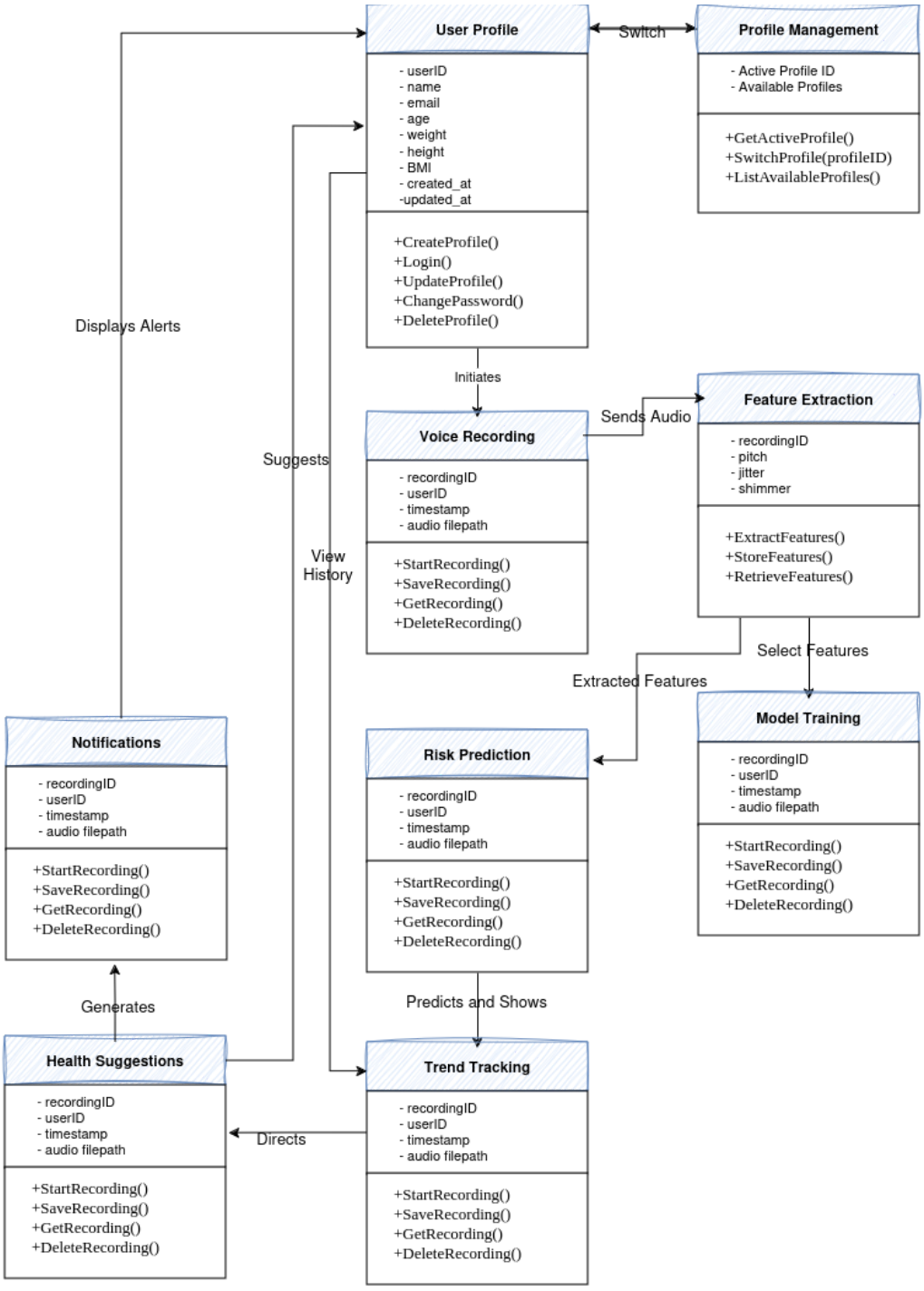


Figure-7: CRC Diagram of AcoustiCare

6. Software Design Architecture

Architectural design is a visual representation in software engineering that outlines the high-level structure and organization of a software system. It focuses on defining the major components or modules of the system, their interactions, and the overall system's architecture.

6.1 Architecture Context Diagram

At the architectural design level, a software architect uses an architectural context diagram (ACD) to model the manner in which software interacts with entities external to its boundaries.

The architectural context diagram for the tool is given below-

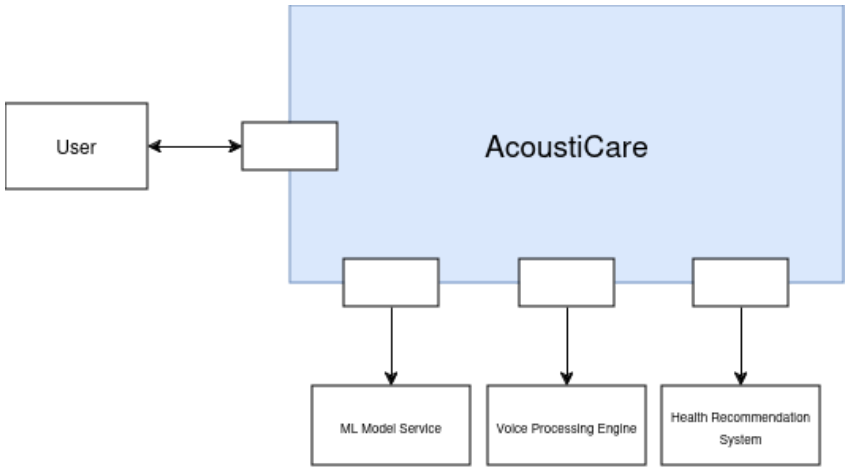


Figure-8: Architectural Context Diagram

The target users for AcoustiCare have similar roles (patients, healthcare providers, individuals concerned about diabetes risk). Due to their similar usage patterns, I combine them into one generic target user group - User. They are both producers (voice recordings, profile data) and consumers (risk assessments, recommendations) of information through the application.

There are three subordinate systems:

1. ML Model Service - Analyzes voice features and predicts diabetes risk
2. Voice Processing Engine - Processes raw voice recordings using Praat libraries
3. Health Recommendation System - Generates personalized health suggestions

These subordinate systems provide the essential processing required for voice-based diabetes risk assessment and monitoring.

6.2 Top Level Components (TLC)

Overall architectural structure with top level components is illustrated below. Note that, Top Level Components diagrams show a very high level view of our system. It is further refined into architecture in the following sections. I keep the top level diagrams to 3 independent responsibilities due to the inherent smaller size of the system and for ease of understanding the system at higher level.

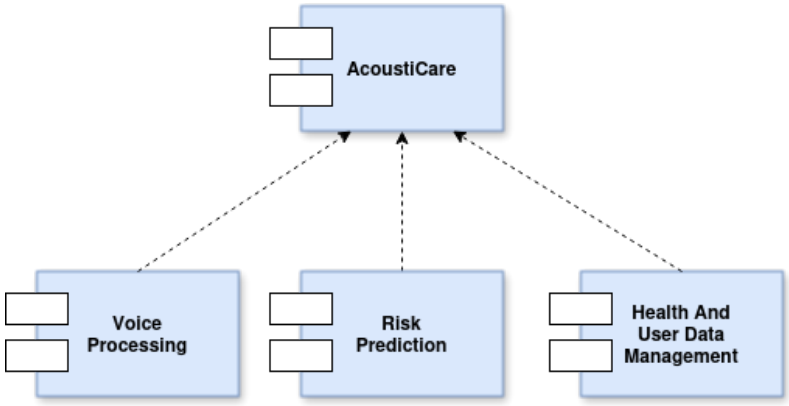


Figure-9: Top Level Components

6.3 Instantiation of the Architecture

The components identified in Section-6.2 are broad. They need to be refined into smaller components. And just a collection of components doesn't create a meaningful design. So I apply architecture on the components to make a maintainable collection of components. The architecture is applied to a specific problem with the intent of demonstrating that the structure and components are appropriate

Component: Voice Processing

The Pipe and Filter architecture is a software design pattern that organizes a system as a series of processing elements (filters) connected by communication channels (pipes). It is particularly useful for designing applications that involve processing streams of data in a linear and sequential manner. Each filter performs a specific task on the incoming data and passes the results to the next filter through the pipes, creating a data flow pipeline.

- **Filter:** All processes that contain a significant general algorithm that can be reused in other tasks. The filters can be re-ordered to solve different problems.
- **Pipe:** Specific tasks that are required to communicate between the filters.

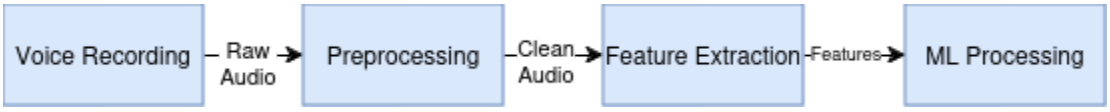


Figure-10: Architecture Instantiation on component Voice Processing

Component: Risk Analysis

The Main Program/Subprogram architectural pattern is a hierarchical structure where a main program coordinates the overall logic and functionality, and specialized subprograms handle specific tasks or subtasks. This pattern is well-suited for components that have a clear separation of concerns and can be decomposed into a main high-level process and several supporting subprocesses.

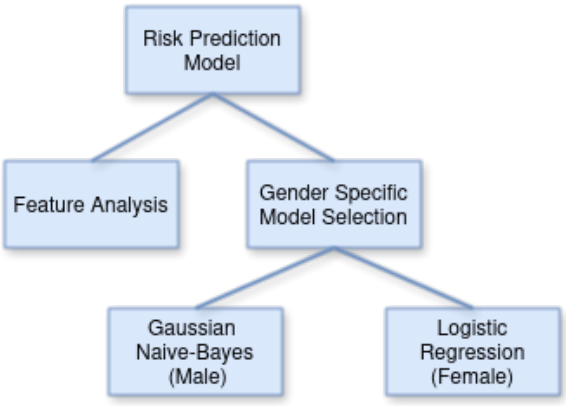


Figure-11: Architecture Instantiation on component Risk Analysis

Component: Health and User Data Management

The Data-Centered architectural pattern places the data store at the center of the system, with other components interacting with the data store to access, manipulate, and store data. This pattern is particularly well-suited for systems where data management and integrity are of critical importance, and where multiple components need to collaborate around a shared data source

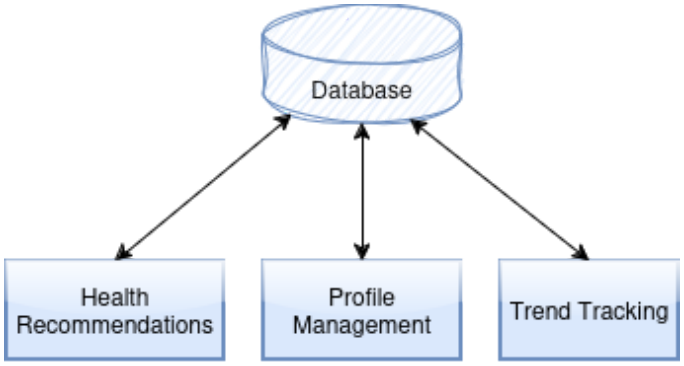


Figure-12: Architecture Instantiation on component Health and User Data Management

Combined Architecture of AcoustiCare: Combining the three TLC into AcoustiCare architecture we get,

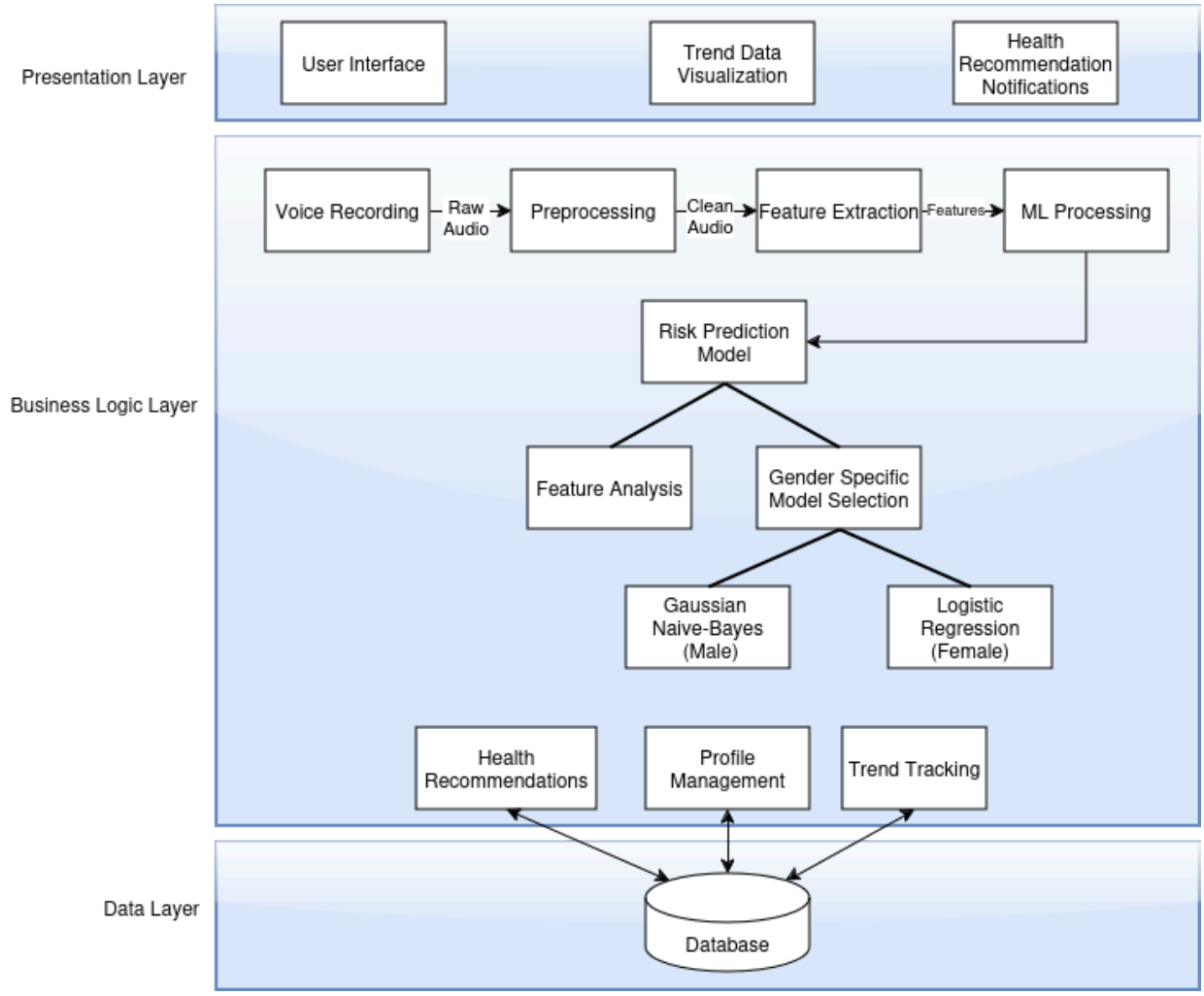


Figure-12: Architecture Instantiation of AcoustiCare

6.4 Mapping Requirements to Software Architecture

The mapping of the derived components with the requirements can be shown as below:

Table-11: Mapping of Components with Requirement

Requirements	**Components**
Users can create profiles, view a list of existing profiles, and manage profile details	Health and Profile Data Management
Users can record their voice using a smartphone microphone	Voice Processing

The app must analyze the recorded audio to extract key acoustic features like pitch, jitter, and shimmer	Voice Processing
Based on the extracted features, the app should predict the risk of T2DM using gender-specific machine learning models	Risk Analysis
Users must receive real-time feedback on their T2DM risk with clear and concise explanations	Risk Analysis
The app should provide users with graphical representations of their T2DM risk trends over time	Risk Analysis
The app should provide personalized health suggestions for lifestyle improvements based on the risk analysis	Health and Profile Data Management
The app should allow users to maintain individual profiles with health data for better tracking and insights	Health and Profile Data Management
The app should generate and deliver real-time notifications to users about their T2DM risk levels and trends	Health and Profile Data Management

7. Preliminary Test Plan

In this chapter, we present a high-level description of the testing goals and a summary of features to be tested for the AcoustiCare application.

7.1. High-Level Description of Testing Goals

The testing goals for AcoustiCare are outlined below:

- To verify that the application meets its functional requirements
- To ensure voice recording works correctly across devices
- To validate acoustic feature extraction (pitch, jitter, shimmer)
- To confirm T2DM risk prediction accuracy based on extracted features
- To verify real-time prediction and feedback functionality
- To minimize false positives/negatives in risk prediction
- To verify accurate trend tracking over time

- To ensure personalized and relevant health suggestions
- To validate seamless profile switching between users
- To verify notification system for risk levels and trends
- To evaluate performance under heavy usage conditions

7.2. Test Cases

- **Test Case 1:** Installation and Setup

  Test Scenario:

  - Install AcoustiCare app
  - Create new user profile

  Expected Outcome:

  - Successful app installation
  - Successful profile setup

- **Test Case 2:** Voice Recording

  Test Scenario:

  - Record voice sample

  Expected Outcome:

  - Clear recording saved without distortion
  - Prompt for next step

- **Test Case 3:** Feature Extraction

  Test Scenario:

  - Extract acoustic features from recording

  Expected Outcome:

  - Numeric values of pitch, jitter, and shimmer metrics

- **Test Case 4:** Risk Prediction

  Test Scenario:

  - Generate T2DM risk prediction from features

  Expected Outcome:

  - Accurate risk assessment considering gender differences

- **Test Case 5:** Real-time Feedback

  Test Scenario:

  - Check prediction feedback display

  Expected Outcome:

  - Display the feedback of prediction result

- **Test Case 6:** Trend Tracking

  Test Scenario:

  - Record samples over multiple days

  Expected Outcome:

  - Accurate visualization of risk trends

- **Test Case 7:** Health Suggestions

  Test Scenario:

  - Generate lifestyle recommendations

  Expected Outcome:

  - Personalized suggestions based on risk trends

- **Test Case 8:** Profile Management

  Test Scenario:

  - Switch between user profiles

  Expected Outcome:

  - Correct profile data loading

- **Test Case 9:** Notification System

  Test Scenario:

  - Trigger notification to audio record voice on the set time

  Expected Outcome:

  - Displays notification on the set time

- **Test Case 10:** Performance Testing

  Test Scenario:

  - Multi-user testing with frequent assessments

Expected Outcome:

- Smooth performance without lag

- **Test Case 11:** Data Consistency

  Test Scenario:

  - Test data storage and retrieval

  Expected Outcome:

  - Data integrity maintained during operations

Conclusion

In this report, I have discussed all the necessary technical details for developing **AcoustiCare: An Application for Acoustic Analysis and Prediction of Type 2 Diabetes Using Smartphone-Recorded Voice Data**. I have covered the scope, assumptions, and usage scenarios. Additionally, scenario-based modeling, data modeling, and class-based modeling have been performed. The architectural design of the application has been discussed, providing a clear view of how various components interact.

I have also presented a high-level description of testing goals, along with a summary of the features to be tested. Detailed test cases have been outlined to ensure the proper functionality of voice recording, feature extraction, risk prediction, trend tracking, and notification management. The goal of these tests is to verify the accuracy and reliability of the application in providing real-time feedback for Type 2 Diabetes risk.

Although **AcoustiCare** offers an innovative, non-invasive method for early detection of Type 2 Diabetes, there is potential for further improvements. Future enhancements could include adding support for additional health conditions, integrating more advanced machine learning models, and extending the application to support multiple languages.

With this document, I have strived to clearly communicate the workflow and technical design of the **AcoustiCare** application. I hope this report provides the reader with a comprehensive understanding of the technical details involved in developing this tool.

References:

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